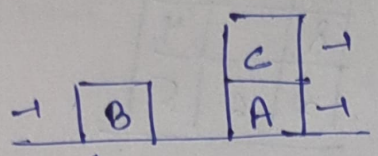
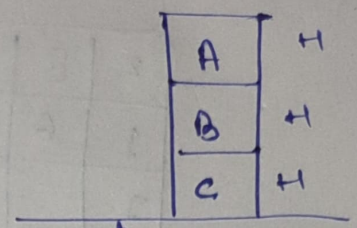
FOAI - Assignment 4

Q.1 Block World Problem



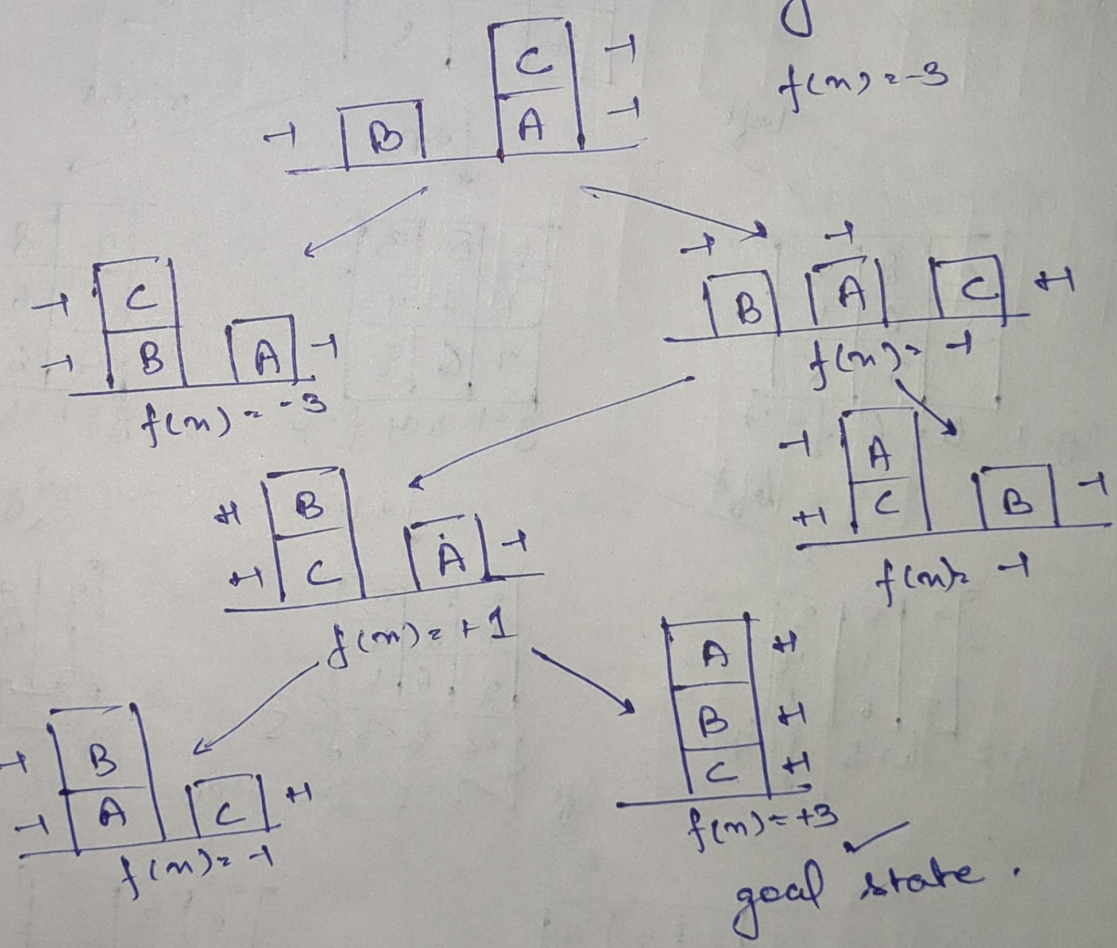
$f(n) = -3$



$f(n) = +3$

Allowed Moves: stack(x, y), lift(x), putg(x)
unstack(x, y)

global heuristic $f^n \longrightarrow$ +1 for every block resting on correct bottom.
-1 for every block resting on wrong bottom.



Q.2 8 Puzzle Problem (A* Algorithm)

2	8	3
1	6	4
7		5

→

1	2	3
8		4
7	6	5

$f(n) = g(n) + h(n)$, $g(n) \rightarrow$ cost to reach the node.

$h(n) \rightarrow$ heuristic.

possible moves → up, left, right, down.

2	8	3
1	6	4
7		5

$g = 0$
$h = 8$
$f = 5$

left ↙ | ↓ up | right ↘

2	8	3
1		4
	7	5

$g = 1$ $h = 6$ $f = 7$

2	8	3
1		4
7	6	5

$g = 1$ $h = 4$ $f = 5$

2	8	3
1	6	4
7	8	

$g = 1$ $h = 6$ $f = 7$

left ↙ | ↓ up | right ↘

2	8	3
	1	4
7	6	5

$g = 2$ $h = 5$ $f = 7$

2		3
1	8	4
7	6	5

$g = 2$ $h = 3$ $f = 5$

2	8	3
1	4	
7	6	5

$g = 2$ $h = 5$ $f = 7$

left ↙ | right ↘

	2	3
1	8	4
7	6	5

$g = 3$ $h = 2$ $f = 5$

2	3	
1	8	4
7	6	5

$g = 3$ $h = 4$ $f = 7$

down

1	2	3
7	8	4
	6	5

← down

1	2	3
	8	4
7	6	5

$g = 4$ $h = 1$ $f = 5$

right →

1	2	3
8		4
7	6	5

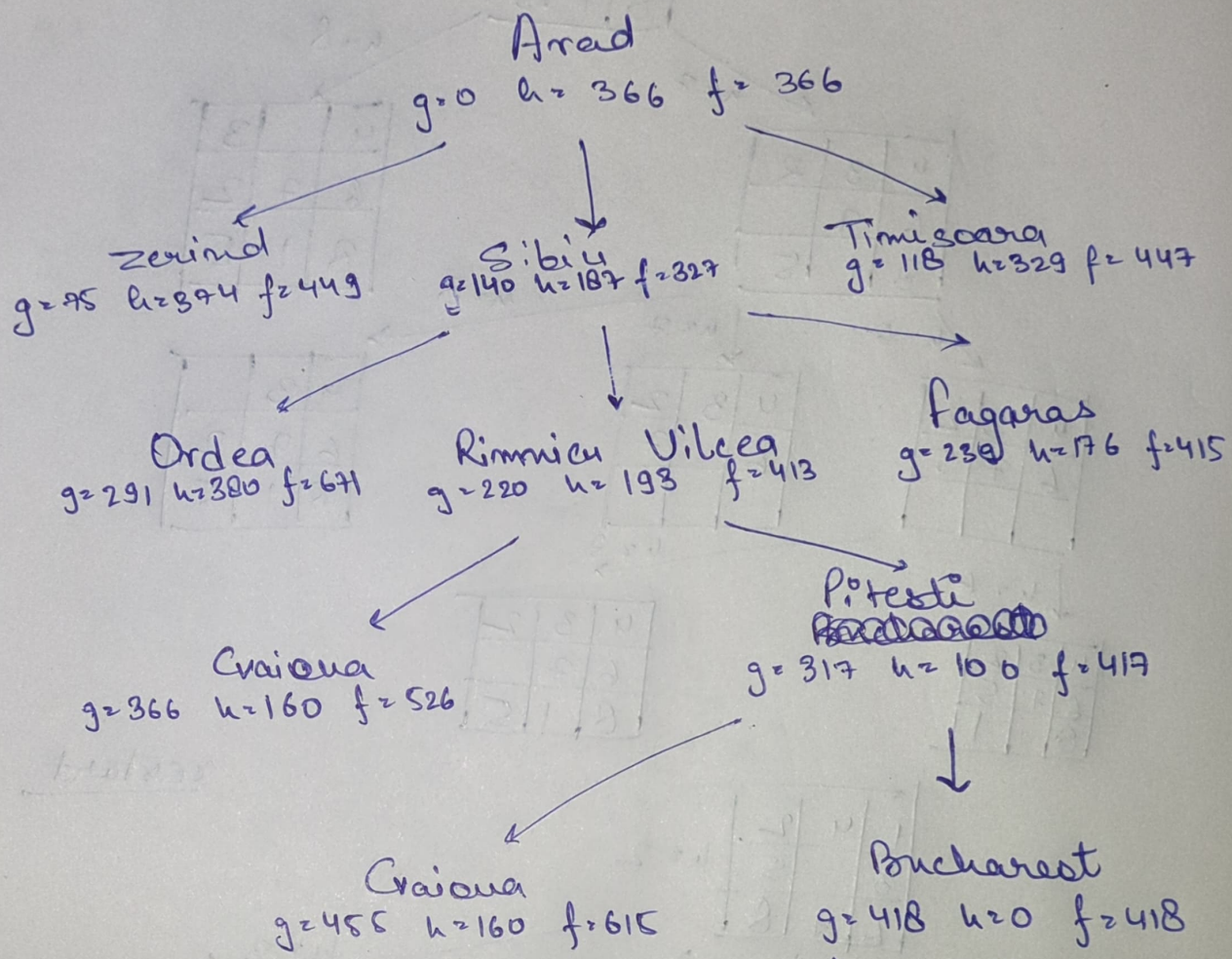
$g = 5$
$h = 0$
$f = 5$

goal state

Q.3 Map of Romania (Informed Search Algorithm).

Start: Arad Goal Node: Bucharest.

$f(n) = g(n) + h(n)$.



Q.4 Hill Climbing Algorithm

Initial

4	3	
6	7	2
8	1	5

goal

	1	2
3	4	5
6	7	8

h=7 h=8

4	3	2
6	7	
8	1	5

4		3
6	7	2
8	1	5

h=6 h=8 h=7

4	3	2
6	7	5
8	1	

4	3	
6	7	2
8	1	5

4	3	2
6		7
8	1	5

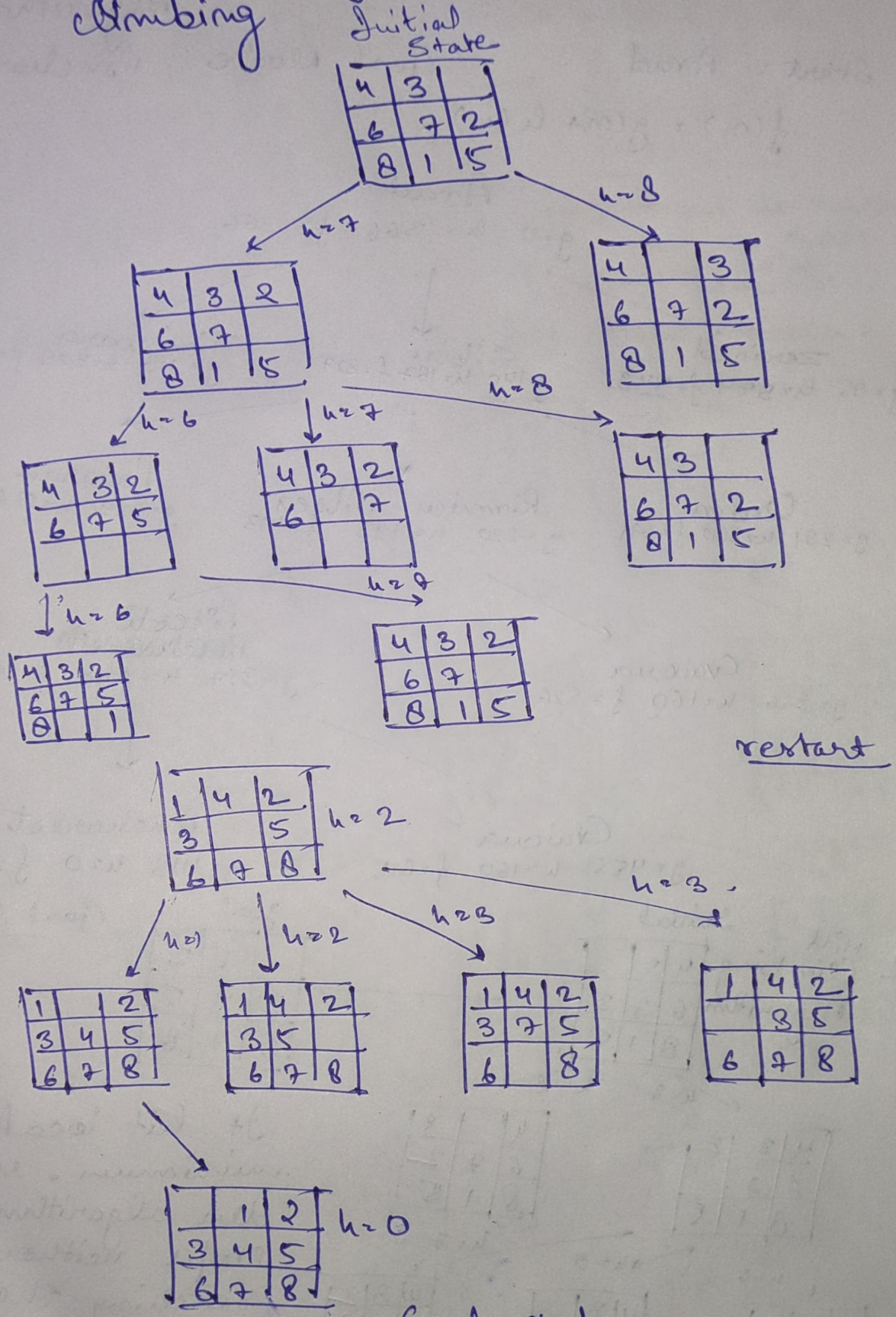
h=6 h=7

4	3	2
6	7	5
8		1

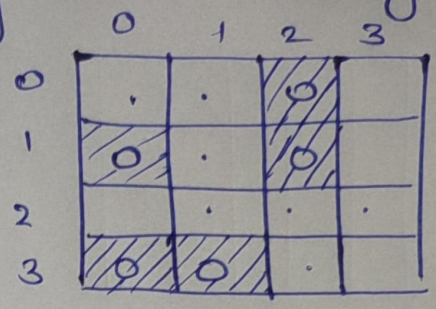
4	3	2
6	7	
8	1	5

It hit local minimum, so the algorithm stops without reaching the goal state.

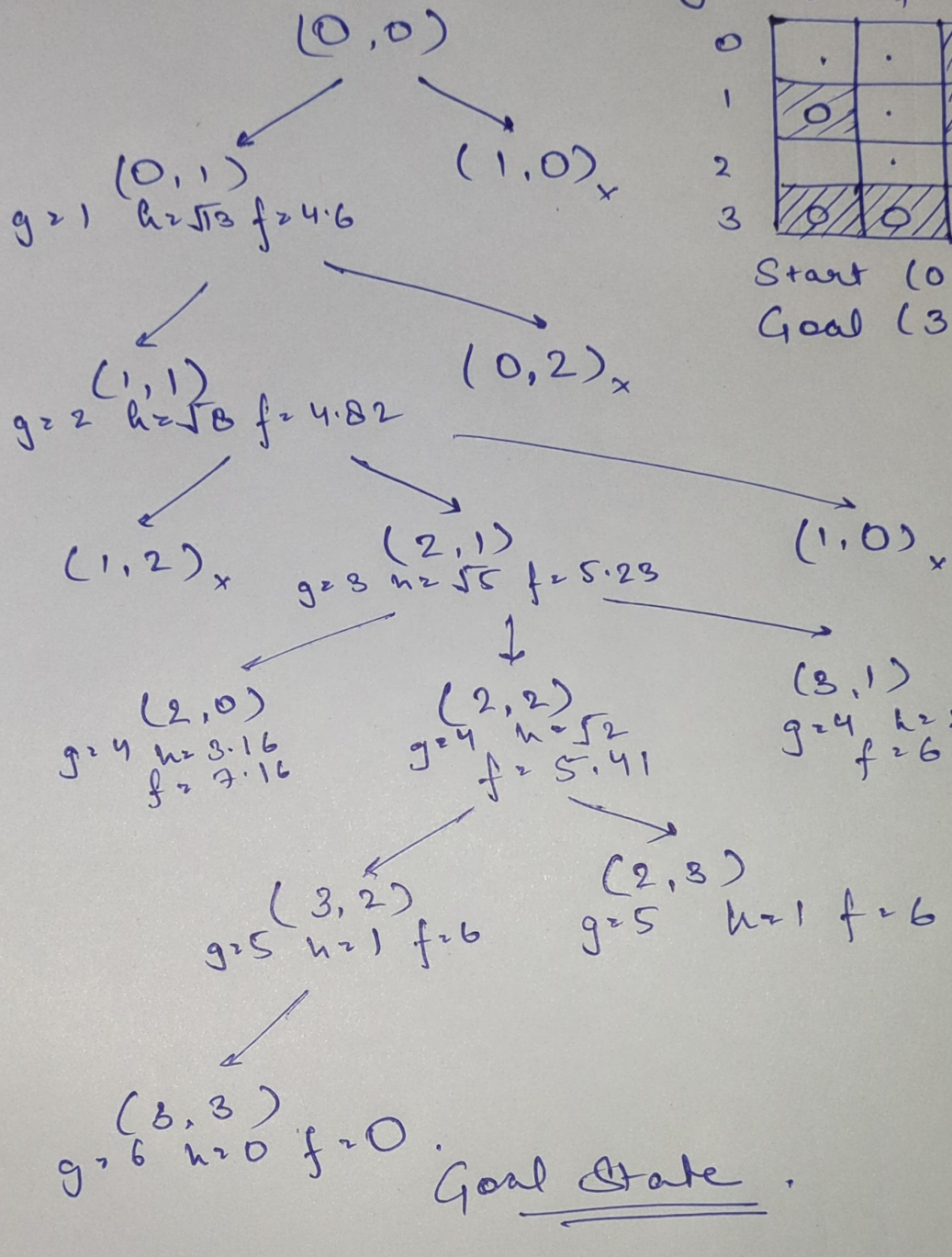
Solving it using Random Restart hill climbing



Q.5 Robot Path Planning in a Maze.



Start (0,0)
Goal (3,3)



Goal State.